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Review article

Physical exercise interventions for health outcomes in children and adolescents with attention deficit hyperactivity disorder: An umbrella review with meta-analysis

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ABSTRACT

Background: Physical exercise is a promising adjunctive intervention for attention deficit hyperactivity disorder (ADHD), but conclusions from previous meta-analyses remain inconsistent.
Objectives: This umbrella review synthesized meta-analytic evidence on the effects of physical exercise interventions on ADHD-related and broader functional outcomes in children and adolescents with ADHD.
Methods: Six databases were searched from inception to March 5, 2026. Eligible meta-analyses evaluating structured physical exercise interventions in children and adolescents with ADHD were included. Random-effects models were used to calculate standardized mean differences (SMDs) with 95% confidence intervals (CIs). Methodological quality and certainty of evidence were assessed using AMSTAR-2 and GRADE, respectively.
Results: Forty-five meta-analyses were included. Physical exercise showed significant beneficial effects on ADHD core symptoms (SMD = 0.50, 95% CI: 0.39 to 0.61) and executive functions (SMD = 0.77, 95% CI: 0.62 to 0.93). Benefits were also observed for mental health, motor skills, and social problems, whereas no significant effects were found for sleep quality, body mass index, or academic achievement. Cognitively engaging exercise and interventions delivered at least three times per week showed significantly larger effects in executive functions. Certainty of evidence was low for ADHD core symptoms, executive functions, mental health, motor skills, and social problems, and very low for other outcomes.
Conclusions: Structured physical exercise may be a useful adjunctive component of multimodal ADHD management, but should not replace established treatments. Given the low to very low certainty of evidence across outcomes, these findings should be interpreted cautiously and require confirmation in further high-quality studies.

1. Introduction

Attention deficit hyperactivity disorder (ADHD) is one of the most prevalent neurodevelopmental disorders in childhood and adolescence, with a global pooled prevalence of approximately 7.2% (Faraone et al., 2024; Wang et al., 2017). Characterized by core symptoms of inattention, hyperactivity, and impulsivity, ADHD often coexists with deficits in executive functions (EFs) (e.g., inhibitory control, working memory, cognitive flexibility) and a range of comorbidities, including mental health issues (anxiety, depression), social difficulties, motor skill impairments, and academic underachievement (Craig et al., 2016; Faraone et al., 2024; Thapar and Cooper, 2016). These multifaceted challenges can persist into adulthood, leading to adverse outcomes such as poor educational attainment, occupational dysfunction, and reduced quality of life (Zhu et al., 2023), placing a substantial burden on individuals, families, and healthcare systems worldwide.

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Current clinical management of ADHD primarily relies on pharmacological interventions (e.g., methylphenidate) and psychological behavioral therapies (e.g., parent training, cognitive-behavioral therapy) (Drechsler et al., 2020; Gosling et al., 2025). While pharmacotherapy can effectively alleviate core symptoms in the short term, it is associated with potential side effects (e.g., sleep disturbances, appetite loss) and long-term safety concerns (Faraone et al., 2024; Li et al., 2024); parent training and cognitive-behavioral therapy, meanwhile, require specialized professionals and structured delivery, making these interventions costly and may not be scalable for larger populations (Zhu et al., 2023). These gaps have prompted a growing emphasis on non-pharmacological, low-cost, and widely accessible interventions. Physical exercise has emerged as a promising candidate: accumulating evidence suggests that structured physical exercise may modulate neurobiological pathways relevant to ADHD (e.g., enhancing dopamine and norepinephrine release, promoting prefrontal cortex development, and reducing neuroinflammation) (He et al., 2025; Huang et al., 2025), thereby exerting beneficial effects on both core symptoms and associated functional impairments, while offering the advantages of low cost, minimal side effects, and broad scalability.

Over the past decade, a surge of systematic reviews and meta-analyses has been conducted to synthesize primary research on physical exercise for ADHD. However, as with all experimental studies, reviews are also prone to inconsistency and methodological biases (e.g., excess significant bias), so that different meta-analyses on identical patients may reach divergent conclusions. For instance, a meta-analysis by Sun et al. (2024) reported a large beneficial effect on ADHD core symptoms (SMD = 1.79, 95% CI: 0.30 to 3.28) (Sun et al., 2024), while a subsequent meta-analysis by Del Carmen et al. (2025) found non-significant effect (SMD = 0.52, 95% CI: −1.16 to 2.19) (Del Carmen et al., 2025), highlighting the inconsistency on conclusions. Additionally, some studies highlight aerobic exercise as most effective for ADHD (Liang et al., 2021), while others emphasize the superiority of cognitively engaging exercise for core symptoms and executive functions improvement (Li et al., 2023). Such conflicting findings hinder efforts to derive firm evidence-based conclusions from individual meta-analyses (Dastamooz et al., 2023). Therefore, some researchers believe these individual meta-analyses exhibit considerable heterogeneity in terms of intervention parameters (e.g., exercise type, intensity, frequency, duration), target populations (age subgroups, ADHD subtypes), and outcome measures, leading to inconsistent conclusions regarding the magnitude and specificity of exercise effects (Zhu et al., 2023).

In recent years, umbrella reviews have provided a useful approach for synthesizing evidence across multiple meta-analyses rather than individual primary studies. Compared with conventional meta-analyses, umbrella reviews can offer a broader overview of the consistency, strength, and limitations of existing evidence, particularly when findings across reviews are heterogeneous or inconclusive (Ioannidis, 2009). Nevertheless, several issues remain insufficiently addressed in the current literature. Existing evidence syntheses have often focused on ADHD core symptoms or selected executive function domains, whereas other outcomes with clinical and developmental relevance, such as mental health, motor skills, social functioning, sleep quality, academic achievement, and body mass index (BMI), have received less integrated evaluation. In addition, recent meta-analyses have not always been examined together with respect to intervention characteristics, including exercise type, intensity, frequency, session duration, and total intervention length. A further issue is that the methodological quality of available meta-analyses and the certainty of evidence across different outcomes require systematic appraisal to support a balanced interpretation of the findings.

Therefore, the present umbrella review had three objectives: to estimate the overall effects of physical exercise interventions on ADHD-related and broader functional outcomes in children and adolescents with ADHD; to examine potential moderators of these effects, including age and exercise characteristics; and to assess the methodological quality of the included meta-analyses and the certainty of evidence for each outcome. Together, these analyses may help clarify the strength and scope of the current evidence and inform the development of more targeted exercise-based interventions for children and adolescents with ADHD.

2. Methods

2.1. Protocol and registration

This umbrella review protocol was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO) prior to the initiation of literature search, with the registration number CRD420251270999. The conduct and reporting of this study strictly adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021) and the Preferred Reporting Items for Overviews of Reviews (PRIOR) guidelines (Gates et al., 2022), ensuring methodological rigor and transparency throughout the entire research process. The PRIOR checklist was provided in the **Supplementary Material S1**.

2.2. Eligibility criteria

The eligibility criteria were developed using the population, intervention, comparison, outcomes and study type (PICOS) framework. The full eligibility criteria were also available on the **Supplementary Material S2**.

Population: children and adolescents formally diagnosed with ADHD according to standardized diagnostic criteria (e.g., Diagnostic and Statistical Manual of Mental Disorders [DSM]-IV to V or International Classification of Diseases [ICD]−10 to 11), with no restrictions on gender or ADHD subtypes (i.e., predominantly inattentive type, predominantly hyperactive-impulsive type, and combined type). The age threshold was set at ≤21 years based on the American Academy of Pediatrics policy statement on the age limit of pediatrics, which notes that the US Department of Health and the Food and Drug Administration identify adolescence as approximately 12 to 21 years of age (US Department of Health and Human Services and Food and Drug Administration, 2003). The same policy statement also indicates that the American Academy of Pediatrics Bright Futures guidelines define adolescence as 11 to 21 years of age, including early adolescence (11–14 years), middle adolescence (15–17 years), and late adolescence (18–21 years) (Hagan et al., 2008). This cutoff was also consistent with several included meta-analyses that defined eligible participants as children and adolescents up to 21 years of age. Importantly, this criterion was not intended to include adult ADHD populations. Studies including both children/adolescents and adults were eligible only when data for children/adolescents or participants aged ≤21 years could be extracted separately; otherwise, they were excluded.

Intervention: chronic physical exercise interventions (≥ 3 weeks in duration) with predefined structure, intensity, frequency, and duration. This threshold was adopted to distinguish repeated exercise interventions from acute or single-session exercise protocols and was consistent with the eligibility criteria used in several included meta-analyses. It was not intended to represent a definitive physiological threshold for long-term adaptation, but rather a pragmatic criterion for identifying structured interventions with repeated exposure. For studies investigating both acute and chronic exercise interventions, only data related to chronic exercise were extracted. In studies combining multiple non-pharmacological interventions, only data exclusively attributable to physical exercise (with independent effect sizes or stratified results) were included. Exercise types included aerobic exercise (e.g., running, swimming, cycling), resistance training, cognitively engaging exercise (e.g., ball games), exergaming, mind-body exercise (e.g., yoga, Tai Chi), and mixed-type exercise. Exercise intensity was categorized as low-intensity physical activity (LPA), moderate-intensity physical activity

(MPA), moderate-to-vigorous intensity physical activity (MVPA), or vigorous-intensity physical activity (VPA) based on definitions in the original studies.

Comparison: control groups included no intervention, usual care, waitlist control, or sedentary activities (e.g., watching videos).

Outcomes: studies must have reported at least one relevant outcome related to ADHD. The primary outcomes were prespecified as ADHD core symptoms and executive functions, because these domains reflect the central clinical and neurocognitive features of ADHD and were the most consistently reported outcomes in the included exercise-intervention literature. ADHD core symptoms included inattention and hyperactivity/impulsivity. Executive functions included inhibitory control, cognitive flexibility, and working memory, measured using validated neurocognitive tasks such as the Stroop task, Wisconsin Card Sorting Task, or N-back task. Secondary outcomes were prespecified as broader functional and developmental outcomes relevant to children and adolescents with ADHD, including mental health (e.g., anxiety, depression, and emotional regulation), motor skills, social problems, sleep quality, BMI, and academic achievement.

Study type: meta-analyses if they analyzed randomized clinical trials (RCTs), non-randomized clinical trials (nRCTs) or quasi-experimental studies published in peer-reviewed journals that reported sufficient statistical details for data analysis. Network meta-analyses were eligible if they provided direct effects for exercise interventions.

The exclusion criteria were as follows: (1) studies that were not meta-analyses (e.g., narrative reviews, conference abstracts, editorials, or study protocols); (2) populations with comorbid neurological disorders (e.g., autism spectrum disorder, cerebral palsy) that could not be stratified from ADHD-only data; (3) interventions that were acute exercise (single-session), unstructured physical exercise, or combined exercise with other non-pharmacological interventions (e.g., cognitive training, behavioral therapy) without independent effect data for exercise alone; (4) studies lacking extractable quantitative data (e.g., unreported effect sizes, missing sample sizes, or ambiguous outcome measures); (5) inaccessible full texts.

2.3. Search strategy

A systematic search was performed across multiple electronic databases (PubMed, Embase, Web of Science, Cochrane, CNKI, and SPORTDiscus) from their inception to March 5, 2026 for identifying all eligible meta-analyses. Key search terms were categorized into four core domains (**Supplementary Material S3**): (1) attention-deficit/hyperactivity disorder (ADHD) and related conditions (e.g., "Attention Deficit Hyperactivity Disorder*", "ADHD", "Hyperkinetic Syndrome", "Hyperactivit*", "Inattenti*", "Impulsivit*"); (2) pediatric and adolescent populations (e.g., "Child", "Adolescent", "Youth*", "Teenager*", "School age", "Preschool*"); (3) physical exercise and related interventions (e.g., "Exercise*", "Physical Activit*", "Aerobic Exercise*", "Resistance exercise", "Exergaming", "Sport*", "Yoga"); and (4) meta-analyses (e.g., "Systematic review", "Meta-analysis", "Evidence synthesis", "Quantitative synthesis"). Boolean operators "AND" and "OR" were used to combine terms within and across domains, respectively, and wildcard symbols (*) were applied to capture variant spellings and word forms. The search strategy was also adapted to the specific requirements of each database. In addition to database searches, manual retrieval of reference lists of included studies and relevant reviews were implemented to enhance search comprehensiveness. No language restrictions were imposed during the initial search to avoid missing potential eligible studies; non-English studies were translated and assessed for eligibility if sufficient methodological and outcome data were available.

2.4. Study selection and data extraction

First, all records retrieved from the electronic databases were imported into EndNote X9 software for management. Duplicate records were removed through a two-step process: automated deduplication using the software's built-in function, followed by manual verification to eliminate residual duplicates. Subsequently, two reviewers (ZFL and LYY) independently screened the titles and abstracts of records against the predefined eligibility criteria. For records with insufficient information to make a definitive judgment, the full texts were retrieved for further assessment. The two reviewers then conducted independent full-text screening to confirm eligibility. Discrepancies arising during either title/abstract or full-text screening were resolved through structured discussion between the two reviewers; if consensus could not be reached, a third senior reviewer (ZM) was consulted to make the final decision. The inter-rater reliability of the screening process was quantified using Cohen's kappa coefficient: fair agreement [0.40–0.59], good agreement [0.60–0.74], and excellent agreement [>0.75] (Sim and Wright, 2005).

A standardized data extraction form was pre-designed based on the PICOS framework and pilot-tested on 5 randomly selected eligible studies to optimize its comprehensiveness and clarity. Modifications were made to the form based on the pilot results to ensure all critical information could be accurately captured. Two independent reviewers (ZFL and WY) extracted data from each eligible meta-analysis using the finalized form. The extracted information included the following core domains: first author, publication year, country of origin, sample size, age range/mean age, gender distribution, exercise type (aerobic, cognitively engaging exercise), length (total weeks), frequency (sessions per week), single-session duration (minutes), intensity (LPA/MPA/MVPA/VPA), type of control group (no intervention, usual care, waitlist, sedentary activity, etc.), primary and secondary outcomes measured. After independent data extraction, the two reviewers cross-validated their extracted data to identify inconsistencies. Discrepancies were resolved through rechecking the original studies and discussion; unresolved disputes were adjudicated by the third senior reviewer (ZM). For studies with missing or ambiguous data (e.g., unreported effect size, incomplete intervention parameters), attempts were made to contact the corresponding authors via email to request supplementary information. If no response was received within one week or the data remained unavailable, the missing information was noted, and the study was retained in the analysis if other critical data were sufficient.

2.5. Methodological quality of meta-analyses

The methodological quality of all meta-analyses was independently assessed by two trained reviewers (ZFL and WY) using the Assessment of Multiple Systematic Reviews tool 2 (AMSTAR-2), a widely validated and recommended tool for evaluating the methodological rigor of systematic reviews and meta-analyses, including those incorporating randomized controlled trials and non-randomized controlled trials (Shea et al., 2017). Each eligible meta-analysis was evaluated against all 16 AMSTAR-2 items, with each item rated as "Yes" (criterion fully met), "No" (criterion not met), or "Unclear" (insufficient information to determine compliance). Critical items (e.g., comprehensive search strategy, risk of bias assessment of primary studies) and non-critical items were distinguished per AMSTAR-2 guidelines. Based on the AMSTAR-2 scoring framework, the methodological quality of each meta-analysis was categorized into four levels: high quality, moderate quality, low quality, and critically low quality. Discrepancies in item interpretation were resolved through group discussion with the third senior reviewer (ZM) to ensure consistent application of the tool across all included studies.

2.6. Certainty of evidence

The certainty of evidence for each outcome was evaluated using the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework (Guyatt et al., 2008). The GRADE ratings were not directly adopted from the included meta-analyses; instead, a

reassessment was conducted for each outcome based on the newly pooled estimates generated in the present umbrella review. For each outcome, the initial certainty rating was determined according to the study designs of the meta-analyses included in the corresponding pooled analysis. Outcomes synthesized exclusively from RCT-only meta-analyses were initially rated as high certainty, whereas outcomes including at least one mixed-design meta-analysis that synthesized both randomized and non-randomized controlled trials were initially rated as moderate certainty. Further downgrading was then considered according to five GRADE domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. To avoid double counting, study design was used to determine the initial certainty rating and was not considered again as an additional downgrading reason.

2.7. Examining the overlap of included meta-analyses

Given that umbrella reviews synthesize evidence from multiple meta-analyses, overlapping of primary studies across included meta-analyses is a common methodological concern. To tackle overlapping citations among the included meta-analyses, we adopted a graphical cross-tabulation approach referred to as a citation matrix (Pieper et al., 2014). The extent of overlap was quantified via an indicator termed the corrected covered area (CCA), which is expressed as a percentage (%). The degree of overlap was classified into four categories based on CCA values: very high (CCA > 15%), high (CCA: 11%–15%), moderate (CCA: 6%–10%), or slight (CCA: 0%–5%) (Comachio et al., 2025). The calculation of CCA was performed using the following formula:

$$CCA = (N - r)/(r \times c - r)$$

c = Number of included reviews; r = Number of publications of primary studies (number of rows); N = Total number of primary study occurrences across all reviews (i.e., counting duplicates).

2.8. Data synthesis

All statistical analyses were performed using R software (version 4.4.0) with the "metafor" and "metaumbrella" packages, adhering to the methodological principles of umbrella reviews. Given the variability in effect size metrics reported across included meta-analyses (e.g., standardized mean difference [SMD], mean difference [MD]), all outcomes were unified to SMD for consistent quantitative synthesis. For studies reporting MD along with sample sizes and confidence intervals (CIs), we first estimated the standard error of the MD using the width of the confidence interval. We then computed the pooled standard deviation based on the sample sizes of the intervention and control groups. Finally, the MD was divided by the pooled standard deviation to obtain the corresponding SMD. Effect sizes were interpreted using Cohen's criteria: SMD = 0.2 (small), 0.5 (moderate), and 0.8 (large) (Cohen, 1992). For timed tests, the direction of effect sizes was harmonized by multiplying values by −1. A random-effects model was employed for all meta-analyses, regardless of heterogeneity magnitude. This choice was justified by inherent between-study variability (e.g., differences in exercise type, intervention duration, outcome assessment tools) and aligns with umbrella review guidelines for synthesizing heterogeneous evidence. The DerSimonian-Laird method was used to estimate between-study variance. Heterogeneity was quantified using the I^2 statistic. I^2 values were interpreted as follows: < 50% (low heterogeneity), 50–75% (moderate heterogeneity), and > 75% (substantial heterogeneity). Sensitivity analyses were conducted by sequentially excluding each included meta-analysis one at a time, to comprehensively test the robustness of the pooled effect size. We also conducted subgroup analyses for primary outcomes and stratified by key moderators, including age subgroup (children [< 12 years], adolescents [12–21 years]), exercise type (aerobic exercise, cognitively engaging exercise), frequency (< 3 sessions/week, ≥ 3 sessions/week), intensity (LPA, MPA, MVPA, VPA), session duration (< 60 min, ≥ 60 min), length (< 12 weeks, ≥ 12 weeks). Although secondary outcomes such as mental health, social problems, sleep quality, BMI, and academic achievement are clinically meaningful, subgroup analyses for these outcomes were not performed because the available evidence was insufficient for reliable stratified estimates. Forest plots were generated to visualize the pooled effect size of each outcome. A summary chart was constructed to synthesize the overall efficacy of exercise interventions across all outcomes.

3. Results

3.1. Study selection

A comprehensive search of electronic databases was conducted, yielding a total of 507 records. After removing duplicate records using EndNote X9 software and manual verification, 356 unique records were retained for title and abstract screening. During title and abstract screening, 278 records were excluded. The full texts of the 78 provisionally included records were retrieved for further assessment. A total of 33 records were excluded at this stage, primarily due to wrong interventions (n = 17), lack of extractable quantitative data (n = 7), not meta-analyses (n = 6), and mismatched populations (n = 3). Finally, 45 eligible meta-analyses were included in the quantitative and qualitative synthesis. The inter-rater agreement for study selection was quantified using Cohen's kappa coefficient; for title and abstract screening, the kappa score was 0.86 (95% CI: 0.80 to 0.92), indicating excellent agreement between the two reviewers (**Supplementary Material S4**). The agreement remained excellent during full-text review, with a kappa score of 0.87 (95% CI: 0.76 to 0.98) (**Supplementary Material S4**). Fig. 1.

3.2. Characteristics of included meta-analyses

A total of 45 eligible meta-analyses were included in this umbrella review, covering a broad range of research on physical exercise interventions for children and adolescents with ADHD. The key characteristics of these included meta-analyses were summarized in Table 1, with additional distributions of country, outcome focus, and methodological quality illustrated in Fig. 2**A-C**. The included meta-analyses were conducted across 9 countries, with a predominance of studies from China (73.4%), followed by Spain (6.7%) and Korea (6.7%), and Japan, Germany, Australia, the USA, the UK, and the Netherlands (each 2.2%) (Fig. 2**A**). The included meta-analyses were published between 2015 and 2025, with 93.3% (n = 42) published after 2021, indicating a growing focus on exercise-based interventions for ADHD in recent years. Among the 45 meta-analyses, 23 (51.1%) synthesized data from both randomized controlled trials and non-randomized controlled trials, 22 (48.9%) focused exclusively on randomized controlled trials.

All included meta-analyses focused on children and adolescents aged ≤ 21 years, with a focus on 5 to 18 years across studies. No meta-analysis imposed restrictions on gender or ADHD subtypes (predominantly inattentive, predominantly hyperactive-impulsive, combined type), with all studies reporting mixed-gender populations. The included meta-analyses covered diverse physical exercise interventions, including aerobic exercise, cognitively engaging exercise. Regarding exercise parameters, the frequency of interventions ranged from 1 to 7 sessions per week (median: 3 sessions/week), single-session duration varied between 15 and 180 min (median: 60 min/session), and total intervention duration spanned 3 to 78 weeks (median: 12 weeks); for intensity, MPA and MVPA were the most commonly reported. The included meta-analyses reported 8 categories of outcomes, with a clear focus on core ADHD-related domains (Fig. 2**B**): executive functions (45.8%) and ADHD core symptoms (22.2%) were the most frequently assessed outcomes; mental health (11.1%), motor skills (8.3%), social problems (5.6%), sleep quality (2.8%), body mass index (2.8%), and academic achievement (1.4%) were also reported.

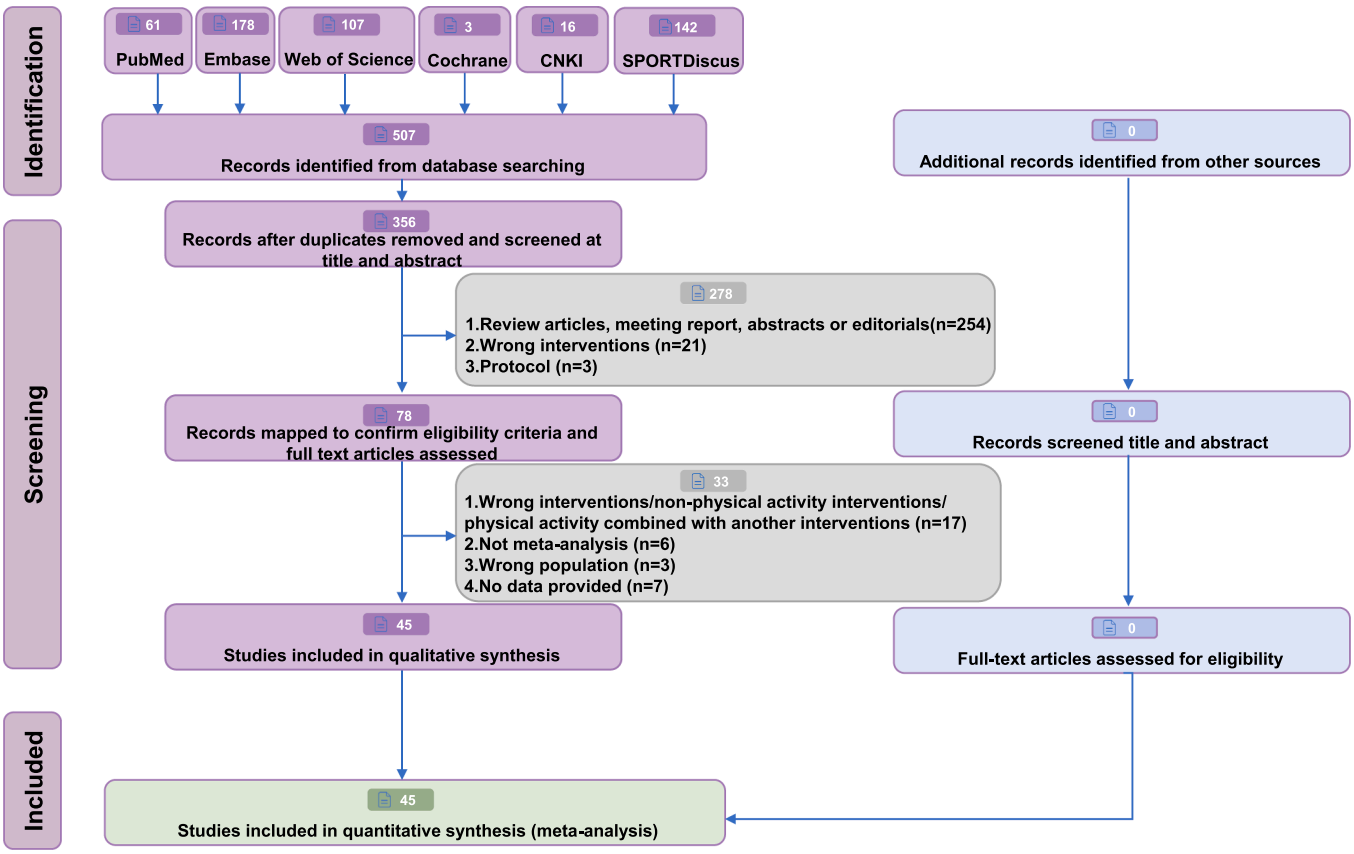


Fig. 1. The flowchart of studies identification and inclusion process.

Methodological quality assessment via the AMSTAR-2 tool revealed that the included meta-analyses were generally of high to low quality (Fig. 2**C**): 53.3% were rated high quality, 28.9% moderate quality, and 17.8% low quality. All included meta-analyses reported the absence of publication bias.

The total number of times that studies appeared in reviews (N) was 254, while the number of rows (primary studies) (r) was 53, with encompassing 45 meta-analysis (c). The CCA was 8.6%, indicating a moderate overlap which was considered methodologically acceptable for umbrella reviews.

3.3. *Effects of physical exercise on ADHD*

3.3.1. *Symptoms*

The pooled effect on overall ADHD core symptoms (combining inattention and hyperactivity/impulsivity) across 33 meta-analyses, with the random-effects model revealing a statistically significant moderate beneficial effect (SMD = 0.50, 95%CI: 0.39 to 0.61, I^2 = 55.2%) (Fig. 3**A**). Fig. 3**B** focused on inattention specifically, synthesizing data from 16 meta-analyses to show a statistically significant moderate effect (SMD = 0.57, 95% CI: 0.39 to 0.74, I^2 = 62.8%); and Fig. 3**C** illustrated the effect on hyperactivity/impulsivity using data from 9 meta-analyses, demonstrating a statistically significant small beneficial effect (SMD = 0.41, 95% CI: 0.20 to 0.62, I^2 = 62.4%).

3.3.2. *Executive functions*

Fig. 4 presented forest plots illustrating the effect of physical exercise on executive functions in children and adolescents with ADHD, stratified by three core subdomains of executive functions. The pooled effect on overall executive functions across 15 included meta-analyses, with the random-effects model showing a statistically significant moderate beneficial effect (SMD = 0.77, 95% CI: 0.62 to 0.93, I^2 = 64.1%) (Fig. 4**A**). Fig. 4**B** focused on cognitive flexibility specifically, synthesizing data from 17 meta-analyses to reveal a statistically significant moderate effect (SMD = 0.66, 95% CI: 0.57 to 0.74, I^2 = 7.7%); and Fig. 4**C** depicted the combined effect on working memory using data from 21 meta-analyses, demonstrating a statistically significant small beneficial effect (SMD = 0.47, 95% CI: 0.37 to 0.56, I^2 = 45.0%); and Fig. 4**D** illustrated the effect on inhibitory control using data from 21 meta-analyses, demonstrating a statistically significant moderate beneficial effect (SMD = 0.65, 95% CI: 0.53 to 0.77, I^2 = 62.5%).

3.3.3. *Mental health*

The pooled effect of physical exercise on mental health in children and adolescents with ADHD, synthesizing data from 13 included meta-analyses. The random-effects model demonstrated a statistically significant moderate beneficial effect (SMD = 0.76, 95% CI: 0.58 to 0.94, I^2 = 24.5%) (**Supplementary Material S5A**).

3.3.4. *Motor skills*

Supplementary Material S5B illustrated the effect of physical exercise on motor skills, drawing on data from 7 meta-analyses. The random-effects model revealed a statistically significant moderate beneficial effect (SMD = 0.71, 95% CI: 0.54 to 0.88, I^2 = 31.9%).

3.3.5. *Sleep quality*

Supplementary Material S5C presented the effect of physical exercise on sleep quality, with only one meta-analysis. The analysis showed a large but non-statistically significant effect (SMD = 2.67, 95% CI: −5.99 to 11.33, I^2 = 96.2%). This non-significant result is likely attributed to the limited sample size from a single study, leading to wide confidence intervals that cross the null line.

Table 1
Characteristics of included meta-analyses.

Study	Country	Included study design	No. of included studies & total sample	Population (gender; age)	Intervention	Comparison	Outcome	AMSTAR category
Bi et al., 2023 (Bi et al., 2023)	China	RCT, nRCT	k = 11 n = 1190	Males and females; 5–18 years	Type=various types Frequency=1–3 sessions/week Time=15–90 min Duration=3–12 weeks Intensity=NR	Routine care, waitlist control	Motor skills	Low
Cerrillo-Urbina et al., 2015 (Cerrillo-Urbina et al., 2015)	Spain	RCT	k = 8 n = 249	Males and females; 6–18 years	Type=Aerobic exercise and Yoga exercise; Frequency=1–5 sessions/week Time=20–90 min Duration=5–20 weeks Intensity=50–75% maximum heart rate or 65–75% of VO_2peak	No physical exercise intervention	Symptoms; Mental health; Executive functions; Social problems	Moderate
Chen et al., 2025 (Chen et al., 2025)	China	RCT, nRCT	k = 29 n = 1231	Males and females; 7–21 years	Type=various types; Frequency=1–7 sessions/week Time=15–180 min Duration=6–72 weeks Intensity=various intensities	Usual care, health education, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	Moderate
Cheng et al., 2025 (Cheng et al., 2025)	China	RCT, nRCT	k = 11 n = 603	Males and females; <18 years	Type=various types; Frequency=1–5 sessions/week Time=40–90 min Duration=5–15 weeks Intensity=NR	No intervention, routine activities, placebo interventions	Executive functions (working memory)	High
Del Carmen et al., 2025 (Del Carmen et al., 2025)	Spain	RCT	k = 20 n = 1645	Males and females; 6–12 years	Type=various types; Frequency=1–5 sessions/week Time=30–90 min Duration=4–15 weeks Intensity=NR	Sedentary attention control program, treatment as usual, waitlist control, watching videos	Symptoms; Executive functions; Sleep quality	Moderate
Fang et al., 2024 (Fang et al., 2024)	China	RCT, nRCT	k = 22 n = 1383	Males and females; 5–18 years	Type=cognitively engaging exercise; Frequency=1–5 sessions/week Time=30–120 min Duration=3–72 weeks Intensity=various intensities	Routine care, waitlist control, watching videos	Symptoms; Executive functions (Inhibitory control, working memory, cognitive flexibility); Motor skills	High
Gonzalez-Devesa et al., 2025 (Gonzalez-Devesa et al., 2025)	Spain	RCT, nRCT	k = 3 n = 131	Males and females; 6–12 years	Type=various types; Frequency=1–3 sessions/week Time=30–60 min Duration=10–12 weeks Intensity=60–100% HR_{max}	No intervention, routine activities	Sleep quality	Moderate
Hamada et al., 2025 (Hamada et al., 2025)	Japan	RCT	k = 2 n = 121	Males and females; < 18 years	Type=exergaming; Frequency=1–3 sessions/week Time=30–60 min Duration=3–8 weeks Intensity=NR	Routine care, waitlist control	Symptoms	High
Huang et al., 2023 (Huang et al., 2023)	China	RCT	k = 22 n = 1309	Males and females; 6–18 years	Type=various types; Frequency=1–7 sessions/week Time=15–180 min Duration=6–72 weeks Intensity=various intensities	Education, no intervention	Symptoms; Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
LaCount et al., 2022 (LaCount et al., 2022)	Korea	RCT, nRCT	k = 11 n = 713	Males and females; < 18 years	Type=various types; Frequency=1–3 sessions/week Time=15–90 min Duration=8–64 weeks Intensity=MPA and MVPA	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control)	Moderate

(*continued on next page*)

Table 1 (*continued*)

Study	Country	Included study design	No. of included studies & total sample	Population (gender; age)	Intervention	Comparison	Outcome	AMSTAR category
Leahy et al., 2025 (Leahy et al., 2025)	Australia	RCT, nRCT	k = 11 n=NR	Males and females; < 18 years	Type=school-based exercise intervention; Frequency=NR Time=NR Duration=NR Intensity=NR	Regular classroom activities	Executive functions; Academic achievement; Mental health	Moderate
Li et al., 2023 (Li et al., 2023)	China	RCT	k = 10 n = 474	Males and females; 6–12 years	Type=various types; Frequency=1–5 sessions/week Time=20–90 min Duration=6–20 weeks Intensity=NR	Education, no intervention	Symptoms	High
Li et al., 2025 (Li et al., 2025)	China	RCT	k = 19 n = 691	Males and females; 6–12 years	Type=various types; Frequency=1–5 sessions/week Time=20–120 min Duration=6–20 weeks Intensity=NR	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Liang et al., 2021 (Liang et al., 2021)	China	RCT, nRCT	k = 15 n = 493	Males and females; 5–18 years	Type=various types; Frequency=2–5 sessions/week Time=30–90 min Duration=6–12 weeks Intensity=various intensities	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	Moderate
Liu et al., 2025 (Liu et al., 2025)	China	RCT	k = 16 n = 685	Males and females; 6–13 years	Type=various types; Frequency=1–5 sessions/week Time=30–90 min Duration=4–12 weeks Intensity=various intensities	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Liu et al., 2025 (Liu et al., 2024)	China	RCT, nRCT	k = 21 n = 926	Males and females; 3–14 years	Type=various types; Frequency=1–6 sessions/week Time=25–120 min Duration=3–24 weeks Intensity=NR	Routine care, no intervention	Executive functions (working memory)	Low
Pan et al., 2025 (Pan et al., 2025)	China	RCT	k = 23 n=NR	Males and females; < 18 years	Type=various types; Frequency=1–3 sessions/week Time=30–90 min Duration=NR Intensity=various intensities	Routine care, education, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Peng et al., 2024 (Peng et al., 2024)	Korea	RCT	k = 7 n = 268	Males and females; < 21 years	Type=mind-body exercise; Frequency=1–3 sessions/week Time=20–90 min Duration=6–24 weeks Intensity=various intensities	Routine care, waitlist control, no intervention	Symptoms; Mental health	High
Qiu et al., 2023 (Qiu et al., 2023)	China	RCT, nRCT	k = 17 n = 471	Males and females; 5–18 years	Type=various types; Frequency=1–5 sessions/week Time=30–90 min Duration=6–72 weeks Intensity=various intensities	Routine care, waitlist control, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Ren et al., 2022 (Ren et al., 2022)	China	RCT, nRCT	k = 11 n = 644	Males and females; < 18 years	Type=various types; Frequency=2–3 sessions/week Time=30–90 min Duration=8–12 weeks Intensity=MPA and MVPA	Routine care, waitlist control, no intervention	Executive functions (Inhibitory control)	Low
Seiffer et al., 2022 (Seiffer et al., 2022)	Germany	RCT	k = 11 n = 416	Males and females; 5–21 years	Type=various types; Frequency=2–3 sessions/week Time=20–80 min Duration=4–12 weeks	Routine care, waitlist control, no intervention	Symptoms; Social problems	High

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Table 1 (*continued*)

Study	Country	Included study design	No. of included studies & total sample	Population (gender; age)	Intervention	Comparison	Outcome	AMSTAR category
					Intensity=MPA and MVPA			
Song et al., 2025 (Song et al., 2025)	China	RCT	k = 17 n = 419	Males and females; 3–18 years	Type=various types; Frequency=2–5 sessions/week Time=30–90 min Duration=5–15 weeks Intensity=various intensities	Sedentary, education, watching videos, no intervention	Executive functions (working memory)	High
Song et al., 2023 (Song et al., 2023b)	China	RCT, nRCT	k = 24 n = 914	Males and females; < 18 years	Type=various types; Frequency=2–3 sessions/week Time=30–90 min Duration=8–12 weeks Intensity=MPA and MVPA	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Song et al., 2024 (Song et al., 2024)	China	RCT	k = 18 n = 830	Males and females; < 18 years	Type=various types; Frequency=1–7 sessions/week Time=30–90 min Duration=2–20 weeks Intensity=MPA and MVPA	Routine care, education	Mental health	High
Song et al., 2023 (Song et al., 2023)	China	RCT	k = 15 n = 592	Males and females; 5–18 years	Type=various types; Frequency=2–7 sessions/week Time=20–90 min Duration=6–72 weeks Intensity=MPA and MVPA	Education, no intervention, sedentary, watching videos, waitlist control	Executive functions (Inhibitory control, working memory, cognitive flexibility)	Low
Sun et al. 2024 (Sun et al., 2024)	China	RCT, nRCT	k = 18 n = 627	Males and females; < 18 years	Type=various types; Frequency=2–3 sessions/week Time=40–90 min Duration=3–12 weeks Intensity=various intensities	Routine care, no intervention, sedentary, watching videos	Symptoms; Executive functions (Inhibitory control, working memory, cognitive flexibility); Mental health; Motor skills; Body mass index	High
Sun et al., 2022 (Sun et al., 2022)	China	RCT	k = 15 n = 734	Males and females; 5–15 years	Type=various types; Frequency=1–5 sessions/week Time=25–90 min Duration=5–78 weeks Intensity=NR	Education, no intervention, sedentary, watching videos, waitlist control	Symptoms; Executive functions; Mental health; Motor skills; Social problems	High
Sung et al., 2022 (Sung et al., 2022)	USA	RCT, nRCT	k = 13 n = 429	Males and females; < 18 years	Type=various types; Frequency=2–5 sessions/week Time=25–120 min Duration=6–15 weeks Intensity=NR	Routine care, no intervention, sedentary	Executive functions	Moderate
Vysniauske et al., 2020 (Vysniauske et al., 2020)	Netherlands	RCT, nRCT	k = 10 n = 300	Males and females; < 18 years	Type=NR; Frequency=NR Time=NR Duration=NR Intensity=MPA and MVPA	No intervention	Executive functions; Motor skills	Moderate
Wang et al., 2025 (Wang et al., 2025a)	China	RCT	k = 7 n = 382	Males and females; 5–18 years	Type=various types; Frequency=2–5 sessions/week Time=30–90 min Duration=8–20 weeks Intensity=NR	Routine care, no intervention	Mental health	High
Wang et al., 2025 (Wang et al., 2025b)	China	RCT	k = 7 n = 312	Males and females; 6–14 years	Type=various types; Frequency=1–3 sessions/week Time=15–90 min Duration=3–12 weeks Intensity=NR	Routine care, no intervention, waitlist control	Motor skills	High
Wang et al., 2023 (Wang et al., 2023)	Korea	RCT, nRCT	k = 11 n = 713	Males and females; < 18 years	Type=various types; Frequency=2–3 sessions/week Time=30–90 min	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control)	Moderate

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Table 1 (*continued*)

Study	Country	Included study design	No. of included studies & total sample	Population (gender; age)	Intervention	Comparison	Outcome	AMSTAR category
					Duration=8–64 weeks Intensity=MPA and MVPA			
Wang et al., 2025 (Wang et al., 2025c)	China	RCT	k = 16 n = 668	Males and females; 6–18 years	Type=aerobic exercise; Frequency=2–3 sessions/week Time=25–120 min Duration=8–78 weeks Intensity=MPA and MVPA	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Welsch et al., 2021 (Welsch et al., 2021)	UK	RCT, nRCT	k = 12 n = 445	Males and females; < 18 years	Type=various types; Frequency=2–3 sessions/week Time=30–90 min Duration=8–78 weeks Intensity=MPA and MVPA	No intervention, waitlist control	Symptoms; Executive functions (Inhibitory control, working memory, cognitive flexibility)	High
Xie et al., 2021 (Xie et al., 2021)	China	RCT, nRCT	k = 23 n = 535	Males and females; < 18 years	Type=various types; Frequency=1–5 sessions/week Time=20–90 min Duration=4–24 weeks Intensity=MPA and MVPA	Routine care, no intervention	Symptoms	Moderate
Xu et al., 2025 (Xu et al., 2025)	China	RCT	k = 19 n = 810	Males and females; < 18 years	Type=various types; Frequency=1–7 sessions/week Time=15–105 min Duration=3–16 weeks Intensity=MPA and MVPA	Sedentary, watching videos, no intervention, waitlist control	Executive functions (Inhibitory control, working memory, cognitive flexibility)	Low
Yang et al., 2024 (Yang et al., 2024)	China	RCT	k = 9 n = 377	Males and females; 6–12 years	Type=aerobic exercise; Frequency=2–3 sessions/week Time=30–90 min Duration=4–12 weeks Intensity=MPA and MVPA	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	Moderate
Yang 2021 (Yang, 2021)	China	RCT, nRCT	k = 9 n = 634	Males and females; < 21 years	Type=various types; Frequency=2–7 sessions/week Time=25–105 min Duration=6–72 weeks Intensity=MPA and MVPA	Sedentary, watching videos, no intervention	Executive functions (Inhibitory control, working memory, cognitive flexibility)	Low
Zang 2019 (Zang, 2019)	China	RCT, nRCT	k = 9 n = 574	Males and females; < 18 years	Type=various types; Frequency=NR Time=NR Duration=NR Intensity=NR	Routine care, no intervention	Symptoms; Mental health; Social problems	Low
Zhang et al., 2025 (Zhang and Li, 2025)	China	RCT	k = 10 n = 526	Males and females; < 14 years	Type=vigorous-intensity exercise; Frequency=2–5 sessions/week Time=20–90 min Duration=6–12 weeks Intensity=VPA	No intervention	Executive functions (Inhibitory control, working memory)	Low
Zhang et al., 2024 (Zhang et al., 2024)	China	RCT	k = 10 n = 683	Males and females; < 18 years	Type=virtual reality-based sports games; Frequency=2–6 sessions/week Time=15–65 min Duration=4–10 weeks Intensity=NR	Routine care, no intervention	Executive functions	Moderate
Zhang et al., 2025 (Zhang et al., 2025)	China	RCT, nRCT	k = 26 n=NR	Males and females; < 18 years	Type=various types; Frequency=2–6 sessions/week Time=10–120 min Duration=4–24 weeks Intensity= MPA and MVPA	Sedentary, watching videos, no intervention	Symptoms; Executive functions (Inhibitory control)	High

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Table 1 (*continued*)

Study	Country	Included study design	No. of included studies & total sample	Population (gender; age)	Intervention	Comparison	Outcome	AMSTAR category
Zhao et al., 2025 (Zhao et al., 2025)	China	RCT, nRCT	k = 26 n = 540	Males and females; 5–18 years	Type=various types; Frequency=1–3 sessions/week Time=30–65 min Duration=6–72 weeks Intensity=various intensities	Routine care, sedentary, watching videos	Symptoms	High
Zhou et al., 2025 (Zhou et al., 2025)	China	RCT	k = 16 n = 546	Males and females; < 18 years	Type=various types; Frequency=2–5 sessions/week Time=30–90 min Duration=6–78 weeks Intensity=various intensities	No intervention, watching videos	Executive functions (Inhibitory control)	High
Zhu et al., 2023 (Zhu et al., 2023)	China	RCT, nRCT	k = 44 n = 1757	Males and females; < 18 years	Type=various types; Frequency=1–7 sessions/week Time=15–90 min Duration=3–72 weeks Intensity=various intensities	Sedentary, watching videos, no intervention, waitlist control	Symptoms; Executive functions (Inhibitory control, working memory, cognitive flexibility)	High

Note. Age ranges are reported according to the eligibility criteria or participant characteristics described in the original meta-analyses. '<14 years,' '<18 years,' and '<21 years' indicate that the corresponding meta-analysis included participants younger than 14, 18, or 21 years, respectively. RCT, randomized controlled trial; nRCT, non-randomized controlled trial; NR, not reported; MPA, moderate physical activity; MVPA, moderate to vigorous physical activity; VPA, vigorous physical activity.

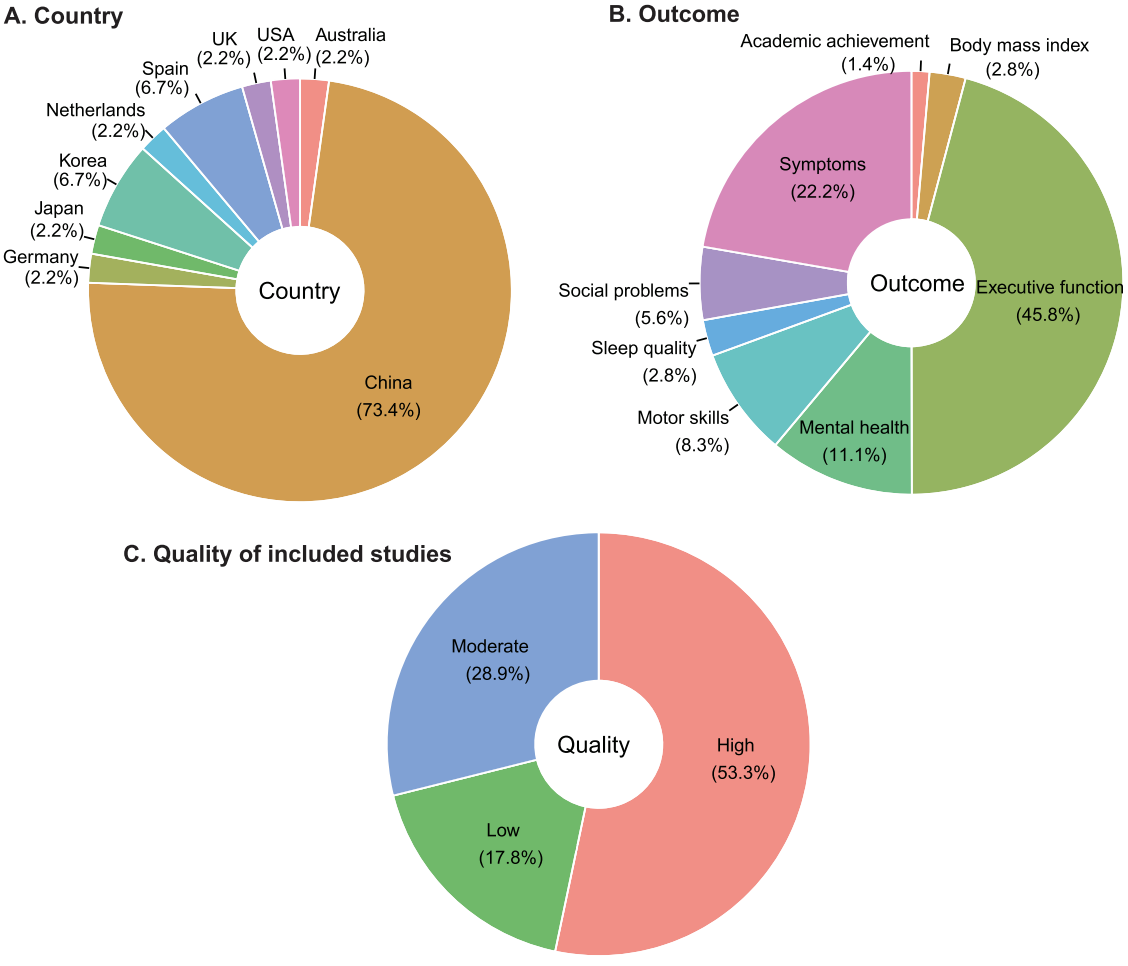


Fig. 2. Distribution diagrams about the country (A), outcome (B), and quality (C) of included studies.

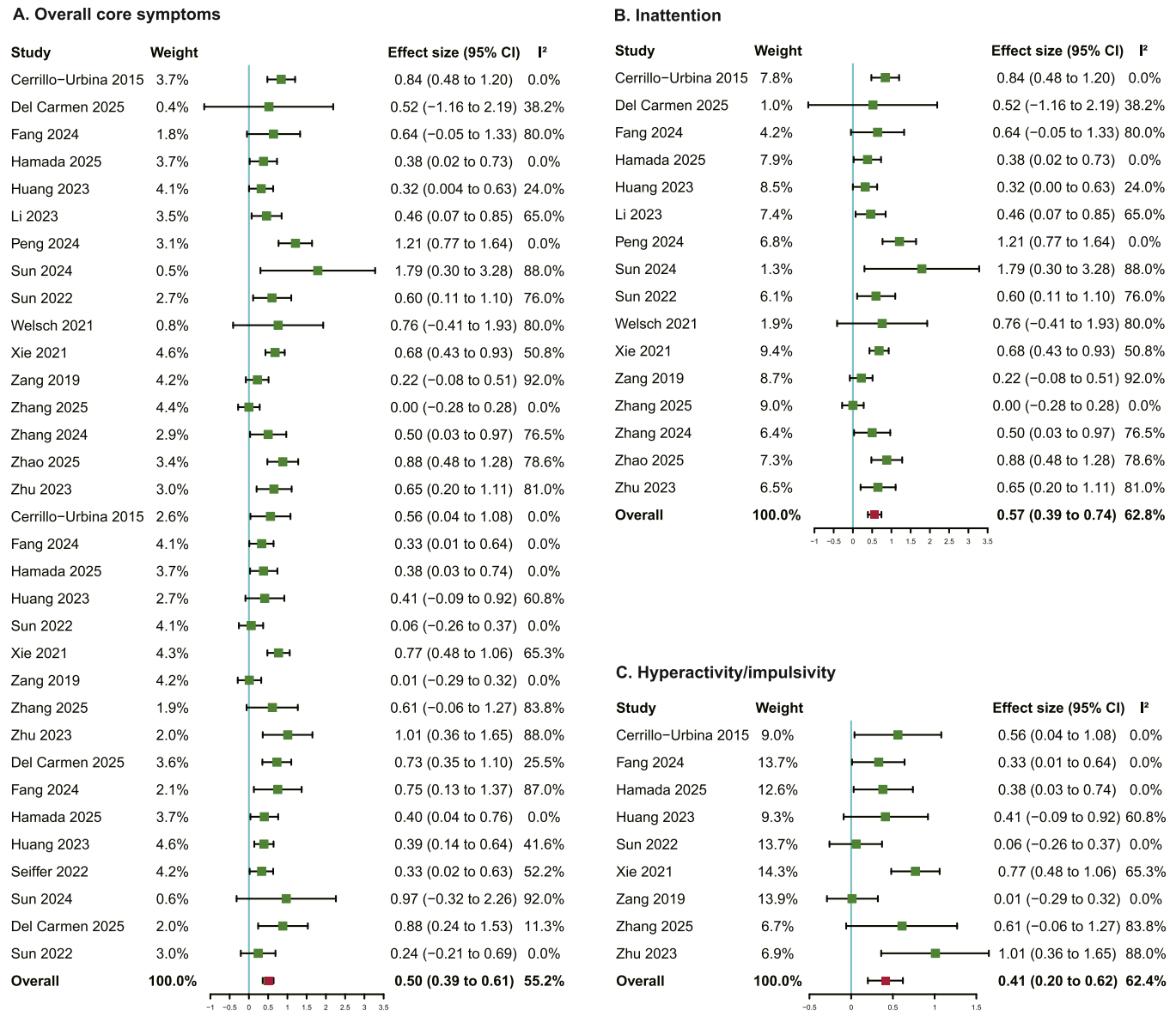


Fig. 3. The forest plot of pooled estimates of each included meta-analysis for core symptoms (A) and different subcomponents (B-C). CI, confidence interval.

3.3.6. *Social problems*

Supplementary Material S5D depicted the pooled effect of physical exercise on social problems, integrating data from 3 meta-analyses. The random-effects model yielded a statistically significant small beneficial effect (SMD = 0.40, 95% CI: 0.16 to 0.64, $I^2 = 0.0\%$).

3.3.7. *Academic achievement*

Supplementary Material S5E summarized the effect of physical exercise on academic achievement, with only one meta-analysis available. The analysis showed a moderate non-statistically significant effect (SMD = 0.56, 95% CI: −0.13 to 1.23), and heterogeneity was not reported. Similar to sleep quality, the non-significant result may be due to the limited evidence from a single study.

3.3.8. *Body mass index*

Supplementary Material S5F illustrated the pooled effect of physical exercise on BMI, synthesizing data from 2 meta-analyses. The random-effects model revealed a small non-statistically significant effect (SMD = 0.19, 95% CI: −0.06 to 0.44, $I^2 = 0.0\%$).

3.4. *Additional analyses*

Sensitivity analyses were performed for the primary outcomes, including ADHD core symptoms and executive functions, to assess the robustness of the pooled effect sizes. Leave-one-out sensitivity analyses, in which each included meta-analysis was sequentially excluded, showed that the pooled effects for both ADHD core symptoms and executive functions remained stable and statistically significant.

Subgroup analyses were conducted to explore the moderating effects of population- and intervention-related factors on ADHD symptoms and executive functions, with results summarized in **Supplementary Material S6**.

3.4.1. *Age*

Regarding age, both outcomes exhibited statistically significant beneficial effects in children (core symptoms: SMD = 0.34, 95% CI: 0.03 to 0.65; executive functions: SMD = 0.55, 95% CI: 0.41 to 0.68) and adolescents (core symptoms: SMD = 0.50, 95% CI: 0.18 to 0.82; executive functions: SMD = 0.65, 95% CI: 0.10 to 1.20). However, the subgroup differences by age were not statistically significant for either ADHD core symptoms (P = 0.485) or executive functions (P = 0.846).

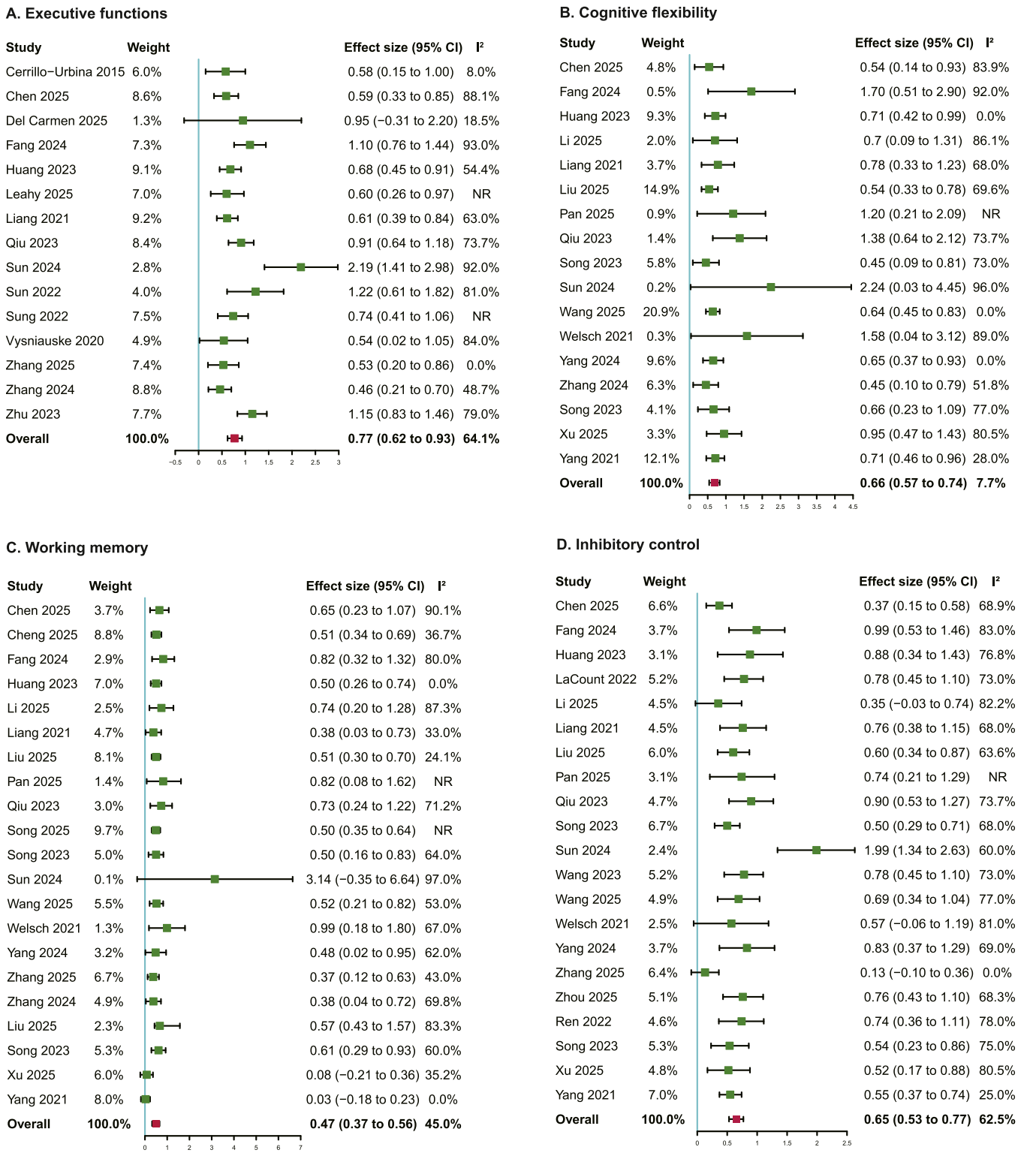


Fig. 4. The forest plot of pooled estimates of each included meta-analysis for executive functions (A) and different subcomponents (B-D). CI, confidence interval; NR, not reported.

3.4.2. *Exercise frequency*

For exercise frequency, interventions delivered at least three sessions per week showed numerically larger effects than those delivered less than three sessions per week for ADHD core symptoms (<3 sessions/week: SMD = 0.42, 95% CI: 0.17 to 0.67; ≥3 sessions/week: SMD = 0.70, 95% CI: 0.40 to 0.99). However, the subgroup difference was not statistically significant for ADHD core symptoms (P = 0.165). For executive functions, interventions delivered at least three sessions per week showed significantly larger effects than those delivered less than three sessions per week (<3 sessions/week: SMD = 0.39, 95% CI: 0.28 to 0.50; ≥3 sessions/week: SMD = 0.64, 95% CI: 0.56 to 0.72), with a significant subgroup difference (P < 0.001).

3.4.3. Exercise intensity

For exercise intensity, MPA showed the largest numerical effect on ADHD core symptoms (SMD = 0.90, 95% CI: 0.32 to 1.48), followed by MVPA (SMD = 0.42, 95% CI: 0.24 to 0.61). LPA (SMD = −0.08, 95% CI: −1.00 to 0.84) and VPA (SMD = 0.19, 95% CI: −0.13 to 0.51) did not show statistically significant effects. However, the subgroup difference by intensity was not statistically significant for ADHD core symptoms (P = 0.141). For executive functions, all intensity subgroups showed statistically significant beneficial effects, including LPA (SMD = 0.46, 95% CI: 0.17 to 0.75), MPA (SMD = 0.58, 95% CI: 0.47 to 0.68), MVPA (SMD = 0.50, 95% CI: 0.40 to 0.59), and VPA (SMD = 0.54, 95% CI: 0.18 to 0.89). The subgroup difference by intensity was not statistically significant (P = 0.630).

3.4.4. Exercise session duration

For session duration, both shorter and longer sessions were associated with significant improvements in ADHD core symptoms (<60 min: SMD = 0.36, 95% CI: 0.13 to 0.60; ≥60 min: SMD = 0.41, 95% CI: 0.15 to 0.68), with no significant subgroup difference (P = 0.777). For executive functions, significant effects were observed for both session durations (<60 min: SMD = 0.58, 95% CI: 0.52 to 0.65; ≥60 min: SMD = 0.72, 95% CI: 0.50 to 0.94). The subgroup difference was not statistically significant (P = 0.327).

3.4.5. Exercise type

Regarding exercise type, both aerobic exercise and cognitively engaging exercise showed significant beneficial effects on ADHD core symptoms. Cognitively engaging exercise showed a numerically larger effect (SMD = 0.65, 95% CI: 0.39 to 0.92) than aerobic exercise (SMD = 0.48, 95% CI: 0.11 to 0.84), but the subgroup difference was not statistically significant (P = 0.499). For executive functions, cognitively engaging exercise showed a significantly larger effect (SMD = 0.64, 95% CI: 0.56 to 0.72) than aerobic exercise (SMD = 0.33, 95% CI: 0.25 to 0.42), with a significant subgroup difference (P < 0.001).

3.4.6. Exercise length

For total intervention length, significant effects on ADHD core symptoms were observed for both shorter and longer interventions (<12 weeks: SMD = 0.32, 95% CI: 0.13 to 0.52; ≥12 weeks: SMD = 0.40, 95% CI: 0.10 to 0.71). The subgroup difference by intervention length was not statistically significant (P = 0.484). For executive functions, significant effects were also observed for both intervention lengths (<12 weeks: SMD = 0.58, 95% CI: 0.48 to 0.63; ≥12 weeks: SMD = 0.60, 95% CI: 0.48 to 0.71), with no significant subgroup difference (P = 0.573).

3.5. Certainty of evidence

As summarized in **Supplementary Material S7** and Fig. 5, the certainty of evidence varied across outcomes. Fig. 5 provided a visual summary of the direction and magnitude of pooled effects together with the corresponding certainty ratings, whereas **Supplementary Material S7** presented the detailed GRADE evidence profile, including the number of included meta-analyses, pooled effect estimates, heterogeneity, initial certainty ratings, downgrading reasons, and final certainty ratings

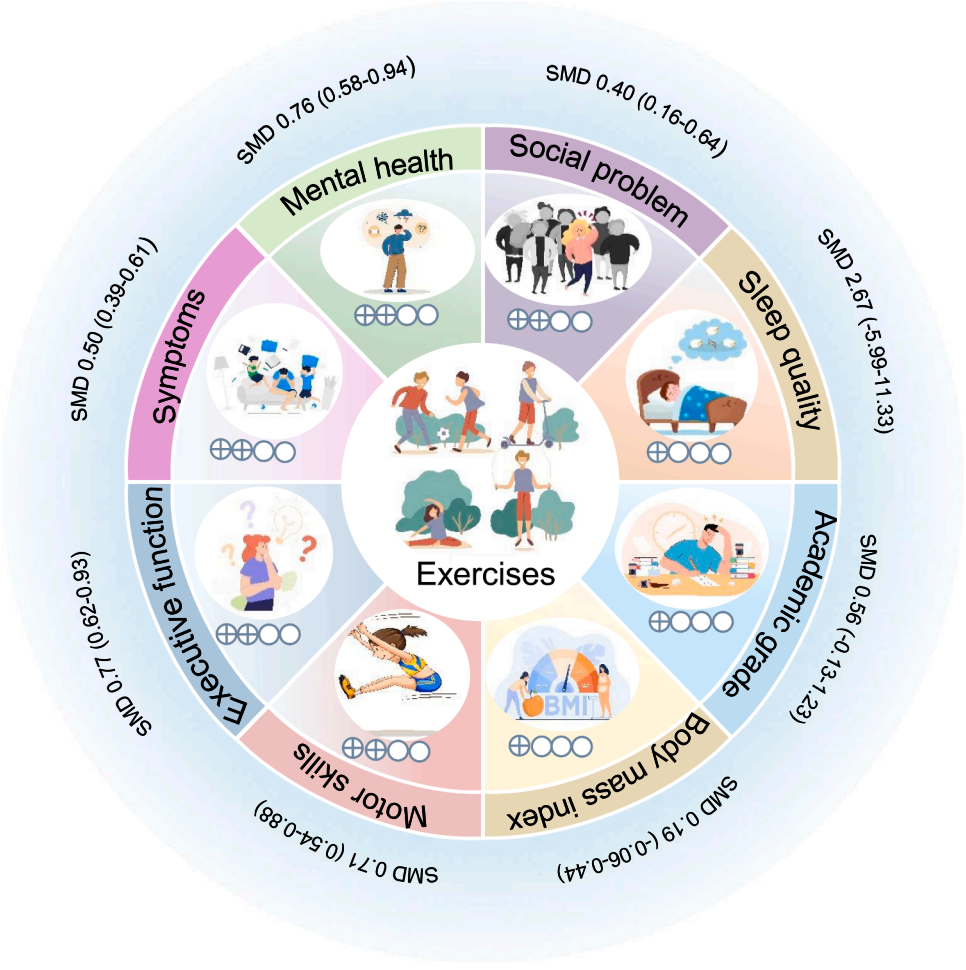


Fig. 5. Summary visualization of the pooled effects of physical exercise interventions across outcomes and their corresponding GRADE certainty ratings. SMD, standardized mean difference.

for each outcome. ADHD core symptoms and executive functions were rated as low-certainty evidence. Although both outcomes showed statistically significant beneficial effects, the certainty was downgraded mainly because the pooled analyses included mixed-design meta-analyses and showed moderate heterogeneity. Mental health and motor skills were also rated as low certainty, largely due to variability in outcome constructs, measurement approaches, and the underlying evidence base. Evidence for social problems was rated as low certainty because, although the pooled effect was statistically significant and heterogeneity was low, the evidence was based on only three meta-analyses. Evidence for sleep quality, BMI, and academic achievement was rated as very low certainty, primarily because these outcomes were supported by only one or two meta-analyses, had wide confidence intervals crossing the null, or showed substantial heterogeneity where applicable. Overall, the findings suggest potential benefits of physical exercise interventions for several ADHD-related and functional outcomes, but the certainty of evidence remains limited and should be interpreted cautiously.

4. Discussion

This umbrella review synthesized evidence from 45 meta-analyses on physical exercise interventions for children and adolescents with ADHD. The findings suggest that structured physical exercise may improve ADHD core symptoms and executive functions, and may also benefit selected broader functional outcomes, including mental health, motor skills, and social problems. In contrast, no statistically significant effects were found for sleep quality, academic achievement, or BMI. Based on the GRADE assessment, however, the certainty of evidence was low for ADHD core symptoms, executive functions, mental health, motor skills, and social problems, and very low for sleep quality, BMI, and academic achievement. Thus, the present findings should be interpreted as supportive but not definitive evidence for the adjunctive role of exercise in ADHD management.

The observed benefits for ADHD core symptoms and executive functions are biologically and clinically plausible. ADHD is characterized not only by behavioral symptoms of inattention, hyperactivity, and impulsivity, but also by deficits in executive control, including inhibitory control, cognitive flexibility, and working memory. Structured exercise may influence these domains through several pathways, including increased arousal regulation, enhanced catecholaminergic activity, improved cerebral blood flow, and neurotrophic processes relevant to prefrontal and fronto-striatal functioning (Benzing and Schmidt, 2017; Herold et al., 2018; Fang et al., 2024; Peng et al., 2024). These mechanisms may partly explain why exercise interventions were associated with improvements in both symptom-level and cognitive outcomes. Nevertheless, the low certainty of evidence indicates that these effects should be interpreted with caution, particularly given the heterogeneity in intervention protocols, participant characteristics, and outcome measures across the included reviews.

A closer examination of the effects on specific executive function subdomains reveals a nuanced pattern. The largest pooled effect sizes were observed for cognitive flexibility and inhibitory control, followed by a small effect on working memory. This hierarchy of responsiveness may reflect the distinct neuroanatomical substrates and cognitive demands engaging by different exercise modalities (Zhu et al., 2023). Cognitively engaging exercise, which showed superior efficacy overall, inherently require rapid task-switching, adaptive rule application, and response inhibition, which processes directly taxing cognitive flexibility and inhibitory control. These activities may preferentially enhance prefrontal cortex (PFC) connectivity and dopamine neurotransmission in associated circuits (e.g., dorsolateral and inferior frontal cortices). The relatively smaller effect on working memory aligns with some prior hypotheses that its neural underpinnings (e.g., dorsolateral PFC and parietal networks) might require more specific or prolonged intervention to manifest change measurable at the behavioral level (Song et al., 2025; Zhu et al., 2023). Furthermore, methodological heterogeneity in working memory assessment (e.g., simple vs. complex span tasks) across primary studies could have contributed to a more attenuated pooled estimate.

The subgroup findings should be interpreted as exploratory, but they provide useful clues regarding intervention design. For ADHD core symptoms, no statistically significant subgroup differences were observed across age, exercise frequency, intensity, session duration, exercise type, or intervention length, suggesting that current evidence is not sufficient to identify a clearly superior exercise parameter for symptom reduction. For executive functions, however, significant subgroup differences were found for exercise frequency and exercise type. Interventions delivered at least three sessions per week showed larger effects than less frequent interventions, which aligns with principles of neuroplasticity, indicating that regular stimulation may be necessary to induce and maintain neural adaptations (Li et al., 2025; Christiansen et al., 2019). Cognitively engaging exercise also showed larger effects than aerobic exercise, suggesting that concurrent cognitive-motor demands may be particularly relevant for executive function improvement. However, because these subgroup analyses were based on aggregate data from included meta-analyses, these findings should not be interpreted as definitive exercise prescriptions.

The findings for broader functional outcomes were more mixed. Improvements in mental health may be related to both biological and psychosocial mechanisms, including stress regulation, increased self-efficacy, better emotional regulation, and enhanced social participation (Wong et al., 2026). Improvements in motor skills are also plausible, because repeated physical practice directly targets coordination, balance, and movement control, while improved attention and executive control may further support motor learning (Bi et al., 2023). The observed benefit for social problems may reflect the structured and interactive nature of many exercise programs, particularly those involving group activities, rules, cooperation, and peer engagement. These findings suggest that exercise may have effects extending beyond core ADHD symptoms, but the low certainty of evidence means that these broader benefits remain provisional. In contrast, the absence of statistically significant effects on sleep quality, academic achievement, and BMI may reflect a mismatch between general exercise interventions and the mechanisms required to change these outcomes. Sleep is influenced by multiple factors, including bedtime routines, circadian rhythm, medication use, family environment, and comorbid emotional symptoms. Academic achievement depends not only on attention and executive function, but also on curriculum exposure, learning support, school context, and long-term educational engagement. BMI is affected by diet, total energy expenditure, growth stage, metabolic factors, and intervention duration. Therefore, exercise programs designed primarily to improve ADHD symptoms or cognitive functioning may not be sufficiently targeted or intensive to produce measurable changes in these domains. Moreover, sleep quality and academic achievement were each supported by only one meta-analysis, and BMI by only two meta-analyses, which further limits confidence in these findings. These results should therefore be interpreted as insufficient evidence rather than evidence that exercise has no effect on these outcomes.

Several limitations should be acknowledged. First, the influence of comorbidity could not be fully examined. Although reviews involving comorbid neurological disorders were excluded when ADHD-specific data were unavailable, information on psychiatric, developmental, or behavioral comorbidities was not consistently reported. Given the high comorbidity burden in ADHD, this may have affected intervention responses. Second, the age criterion of $\leq$21 years may have introduced developmental heterogeneity. Although this cutoff was adopted to include late-adolescent populations and to remain consistent with several included meta-analyses, participants aged 18–21 years may differ from younger children and adolescents in symptom profiles, neurodevelopmental stage, exercise capacity, and responsiveness to exercise interventions. More refined developmental subgroup analyses

were not feasible because individual participant data were unavailable and age categories were inconsistently reported across reviews. Third, substantial variability existed in exercise protocols and outcome measures, including exercise type, intensity, frequency, duration, setting, and assessment tools, which may have contributed to heterogeneity and reduced comparability across reviews. Fourth, the evidence base for several secondary outcomes was limited. Sleep quality and academic achievement were each based on only one meta-analysis, and BMI was based on two meta-analyses; none of these pooled effects reached statistical significance. In addition, subgroup analyses were not conducted for secondary outcomes because the number of available meta-analyses was small and intervention characteristics were not consistently reported across reviews, which would have resulted in unstable and potentially misleading subgroup estimates. Finally, although primary study overlap was assessed using the corrected covered area and was considered moderate, overlap may still have influenced the precision and interpretation of pooled estimates. These limitations should be considered when applying the findings to clinical practice or public health recommendations.

5. Conclusion

Structured physical exercise may be considered a promising adjunctive component of multimodal ADHD management for children and adolescents, particularly for improving core symptoms and executive functions. Subgroup findings suggest that cognitively engaging exercise and interventions delivered at least three times per week may be especially relevant for executive function improvement. However, exercise should not be viewed as a replacement for established pharmacological, behavioral, educational, or psychosocial interventions. Given the low to very low certainty of evidence across outcomes, exercise-based interventions should be implemented with appropriate clinical judgment and tailored to individual needs. Further high-quality studies with standardized protocols, clearly defined exercise prescriptions, and longer follow-up are needed to clarify the optimal intervention characteristics and long-term benefits for this population.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, generative AI and AI-assisted language tools were solely used for language polishing, grammar correction, and academic English refinement.

Data availability

Data will be made available to academic researchers following a reasonable request.

CRediT authorship contribution statement

Feilong Zhu: Writing – original draft, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Xi Zhang:** Writing – review & editing, Project administration, Methodology, Data curation. **Yu Wu:** Writing – review & editing, Project administration, Methodology, Data curation. **Yingying Li:** Writing – review & editing, Data curation. **Yuanchun Ren:** Writing – review & editing, Supervision, Software, Project administration, Methodology. **Ming Zhang:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.psychres.2026.117267.

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